SCOUTS BSA TEST LAB SERIES

# PHYSICS



Scouting America

Scouting America Merit Badge Reference Guide: Physics
Req 1: Physics Lab Safety, PPE & Hazard Prevention

Safety. Before conducting any experiments, explain to your counselor the safety precautions and personal protective equipment necessary for working with physics-related equipment.

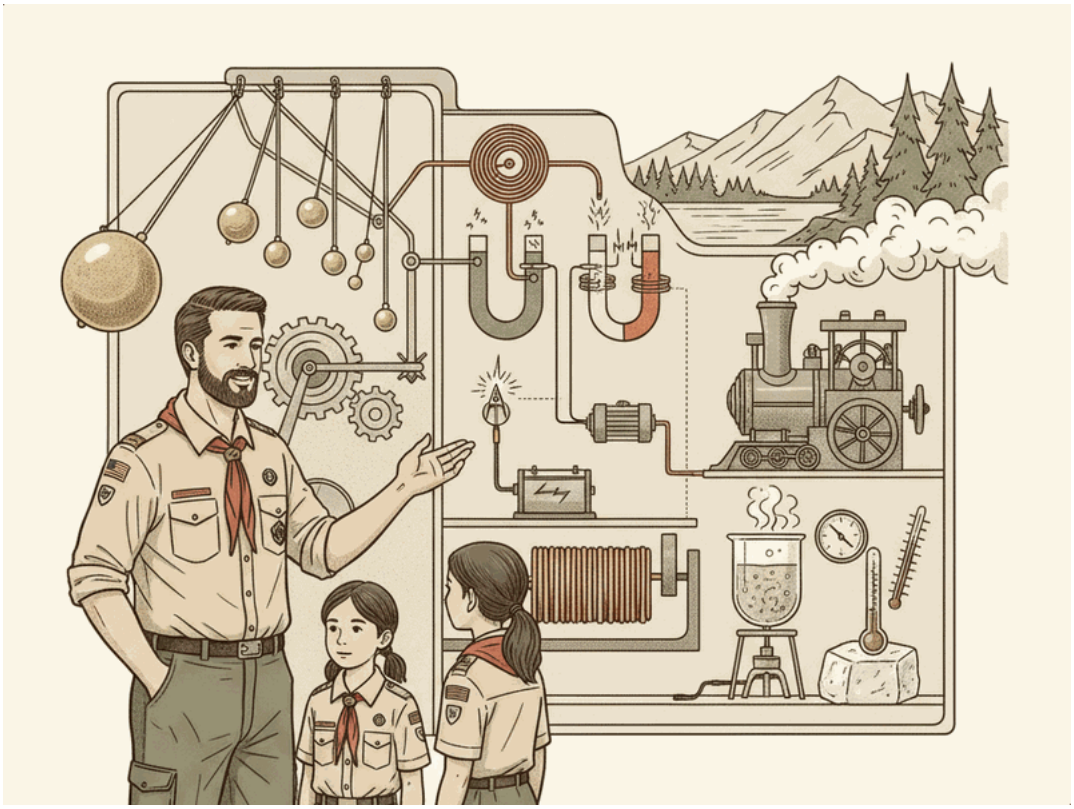
Key Pamphlet Study Points:
- Core Standard 1 (Req 1): Before conducting any experiments, explain to your counselor the safety precautions and personal protective equipment necessary for working.
- Key Terminology & Evidence: Document specific examples, definitions, and field observations for Physics Lab Safety, PPE & Hazard Prevention in your Physics Workbook.
- Safety & Policy Alignment: Follow the BSA Guide to Safe Scouting, Two-Deep Leadership, and the Buddy System during all Physics Req 1 activities.
- Counselor Conference Readiness: Be prepared to explain your findings and personal reflections on Physics Lab Safety, PPE & Hazard Prevention with your Merit Badge Counselor.



Scouting America Merit Badge Reference Guide: Physics
Req 2: Defining Physics & Its Major Scientific Branches

What is Physics? Define physics in your own words and discuss with your counselor various areas of physics.

Key Pamphlet Study Points:
- Core Standard 1 (Req 2): What is Physics? Define physics in your own words and discuss with your counselor various areas of physics.
- Key Terminology & Evidence: Document specific examples, definitions, and field observations for Defining Physics & Its Major Scientific Branches in your Physics Workbook.
- Safety & Policy Alignment: Follow the BSA Guide to Safe Scouting, Two-Deep Leadership, and the Buddy System during all Physics Req 2 activities.
- Counselor Conference Readiness: Be prepared to discuss your findings and personal reflections on Defining Physics & Its Major Scientific Branches with your Merit Badge Counselor.



Scouting America Merit Badge Reference Guide: Physics
Req 3: Fundamental Concepts: Force, Motion, Energy & Waves

Fundamentals of Physics. Explain these key physics concepts: force, motion, energy, and waves. Provide real-world examples demonstrating each principle.

Key Pamphlet Study Points:
- Core Standard 1 (Req 3): Fundamentals of Physics.
- Core Standard 2 (Req 3): Explain these key physics concepts: force, motion, energy, and waves.
- Core Standard 3 (Req 3): Provide real-world examples demonstrating each principle.
- Counselor Conference Readiness: Be prepared to explain your findings and personal reflections on Fundamental Concepts: Force, Motion, Energy & Waves with your Merit Badge Counselor.



Scouting America Merit Badge Reference Guide: Physics
Req 4: Hands-On Physics Experiments & Data Analysis

Hands-On Experimentation. Conduct THREE physics experiments, selecting from the following options. Each experiment must include: a hypothesis, materials used, safety precautions, observations, and conclusions based on results. Explain how each experiment demonstrates key physics principles. For each experiment, be sure to try variations/variables to test your hypothesis. (A) Option A: Newton?s Laws in Action: Balloon-Powered Car. Make and test a balloon-powered car to show how force and motion interact. (B) Option B: Momentum: Stacked Ball Drop. Drop different-sized stacked balls to show momentum. (C) Option C: Centrifugal Force. Make spinning toys to demonstrate centrifugal force. (D) Option D: Centripetal Force. Swing a water bucket or tray filled with water glasses to demonstrate centripetal force. (E) Option E: Balance. Explore center of gravity, inertia, and harmonic motion using a Tweety Bopper. (F) Option F: Visualize Sound. Visualize Chladni patterns using a tonoscope. (G) Option G: Cartesian Diver. Demonstrate buoyancy and the ideal gas law with Cartesian divers. (H) Option H: Air Pressure: Bernoulli?s principle and the Coand? Effect. Balance different objects in a stream of air. (I) Option I: Electromagnet. Make an electromagnet and explore how to increase its strength.

Key Pamphlet Study Points:
- Physics Sub-Req 4A: Option A: Newton?s Laws in Action: Balloon-Powered Car. Make and test a balloon-powered car to show how force and motion interact.
- Physics Sub-Req 4B: Option B: Momentum: Stacked Ball Drop. Drop different-sized stacked balls to show momentum.
- Physics Sub-Req 4C: Option C: Centrifugal Force. Make spinning toys to demonstrate centrifugal force.
- Physics Sub-Req 4D: Option D: Centripetal Force. Swing a water bucket or tray filled with water glasses to demonstrate centripetal force.



Scouting America Merit Badge Reference Guide: Physics
Req 5: Physics in Daily Life, Space, Climate & Scouting

Physics All Around You. Discuss with your counselor: (a) Physics in Everyday Life. Describe FIVE ways physics is applied in daily life, such as in transportation, sports, or household appliances. (b) Physics in Space Exploration. Describe how physics is used in space travel, satellite technology, or planetary exploration. (c) Physics in the Environment. Explain how physics contributes to renewable energy, climate science, or environmental conservation. (d) Physics in Scouting. Explain how physics is found in Scouting, such as knot tying, fire building, cooking, or pioneering.

Key Pamphlet Study Points:
- Physics Sub-Req 5a: Physics in Everyday Life. Describe FIVE ways physics is applied in daily life, such as in transportation, sports, or household appliances.
- Physics Sub-Req 5b: Physics in Space Exploration. Describe how physics is used in space travel, satellite technology, or planetary exploration.
- Physics Sub-Req 5c: Physics in the Environment. Explain how physics contributes to renewable energy, climate science, or environmental conservation.
- Physics Sub-Req 5d: Physics in Scouting. Explain how physics is found in Scouting, such as knot tying, fire building, cooking, or pioneering.



Scouting America Merit Badge Reference Guide: Physics
Req 6: Frontiers of Physics: Quantum, Space & New Theories

The Future of Physics. Physicists try to explain the universe and apply these concepts to novel technologies. Discuss one option with your counselor: (A) Option A: Unsolved Mysteries of Physics. Discuss the following with your counselor: (a) Dark Matter & Dark Energy. Scientists know it exists?but what is it? (b) Black Holes & Wormholes. Can we harness their energy or travel through them? (B) Option B: Quantum Mechanics & Computing. Discuss the following with your counselor: (a) Quantum Computers. How can super fast computers be made of atoms instead of microchips? (b) Quantum Entanglement. Can we communicate instantly across space? (C) Option C: Space Exploration & Astrophysics. Discuss the following with your counselor: (a) Colonizing Mars. How can physics help humans live on other planets? (b) Interstellar Travel. Can we reach other stars with warp drives or solar sails? (D) Option D: Advanced Theories & Future Discoveries. Discuss the following with your counselor: (a) Time Travel. Is it possible under extreme conditions? (b) Artificial Gravity. Can we create Earth-like gravity in space stations?

Key Pamphlet Study Points:
- Physics Sub-Req 6A: Option A: Unsolved Mysteries of Physics. Discuss the following with your counselor.
- Physics Sub-Req 6A(a): Dark Matter & Dark Energy. Scientists know it exists, but what is it?
- Physics Sub-Req 6A(b): Black Holes & Wormholes. Can we harness their energy or travel through them?
- Physics Sub-Req 6B: Option B: Quantum Mechanics & Computing. Discuss the following with your counselor.



Scouting America Merit Badge Reference Guide: Physics
Req 7: Engineering & Physics Design Challenge

Engineering and Physics Challenge. Solve a simple engineering problem using physics-based reasoning. Define the problem, allowable materials, and safety constraints. You may enlist friends or family in completing your challenge. All plans must be approved in advance by your counselor. Do ONE of the following options: (A) Option A: Fastest Pinewood Derby Car. Build pinewood derby cars using physics principles to get the fastest time on the track. (B) Option B: Rube Goldberg Machine. Build a machine using household materials to perform a simple task. You can use examples of tasks from the annual Rube Goldberg competition, or you can create your own task: (C) Option C: Egg Drop Challenge. Explore the physics of energy and momentum by building a structure using household materials to prevent an egg from breaking when dropped from a significant height. The location of the egg drop must be approved by your counselor. (D) Option D: Water Bottle Rockets. Explore propulsion and Newton?s Third Law. The challenge is to land an air pressured rocket, using water as a propellant, exactly at a specific distance (for example, 70 m) from launch point, three times, with extra points for flight duration. (E) Option E: Design your Own Physics Challenge. Create a physics challenge with your counselor?s approval.

Key Pamphlet Study Points:
- Physics Sub-Req 7A: Option A: Fastest Pinewood Derby Car. Build pinewood derby cars using physics principles to get the fastest time on the track.
- Physics Sub-Req 7B: Option B: Rube Goldberg Machine.
- Physics Sub-Req 7C: Option C: Egg Drop Challenge.
- Physics Sub-Req 7D: Option D: Water Bottle Rockets.



Scouting America Merit Badge Reference Guide: Physics
Req 8: Physics Careers, Research & STEM Pathways

Careers. Do ONE of the following: (a) Explore careers related to this badge. Research one career to learn about the training and education needed, costs, job prospects, salary, job duties, and career advancement. Your research methods may include ? with your parent or guardian?s permission ? an internet or library search, an interview with a professional in the field, or a visit to a location where people in this career work. Discuss with your counselor both your findings and what about this profession might make it an interesting career. (b) Explore how you could use knowledge and skills from this badge to pursue a hobby or healthy lifestyle. Research any training needed, expenses, and organizations that promote or support it. Discuss with your counselor what short-term and long-term goals you might have if you pursued this.

Key Pamphlet Study Points:
- Physics Sub-Req 8a: Explore careers related to this badge.
- Physics Sub-Req 8b: Explore how you could use knowledge and skills from this badge to pursue a hobby or healthy lifestyle.
- Core Standard 1 (Req 8): Do ONE of the following: (a) Explore careers related to this badge.
- Core Standard 2 (Req 8): Research one career to learn about the training and education needed, costs, job prospects, salary, job duties, and career advancement.



Scouting America Merit Badge Reference Guide: Physics
Req 9: Scouting America Test Lab Feedback & Evaluation

Complete the survey below to complete the test lab requirements

Key Pamphlet Study Points:
- Core Standard 1 (Req 9): Complete the survey below to complete the test lab requirements.
- Key Terminology & Evidence: Document specific examples, definitions, and field observations for Scouting America Test Lab Feedback & Evaluation in your Physics Workbook.
- Safety & Policy Alignment: Follow the BSA Guide to Safe Scouting, Two-Deep Leadership, and the Buddy System during all Physics Req 9 activities.
- Counselor Conference Readiness: Be prepared to complete your findings and personal reflections on Scouting America Test Lab Feedback & Evaluation with your Merit Badge Counselor.

